

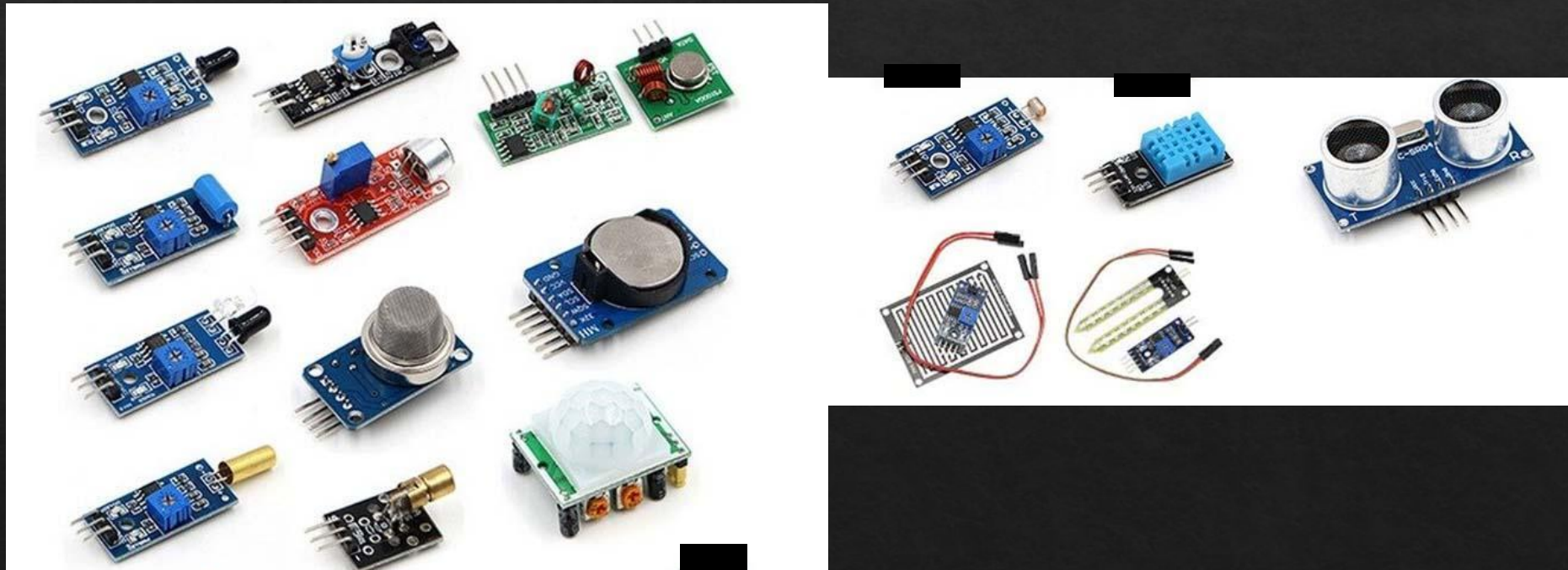


Contents of this Units

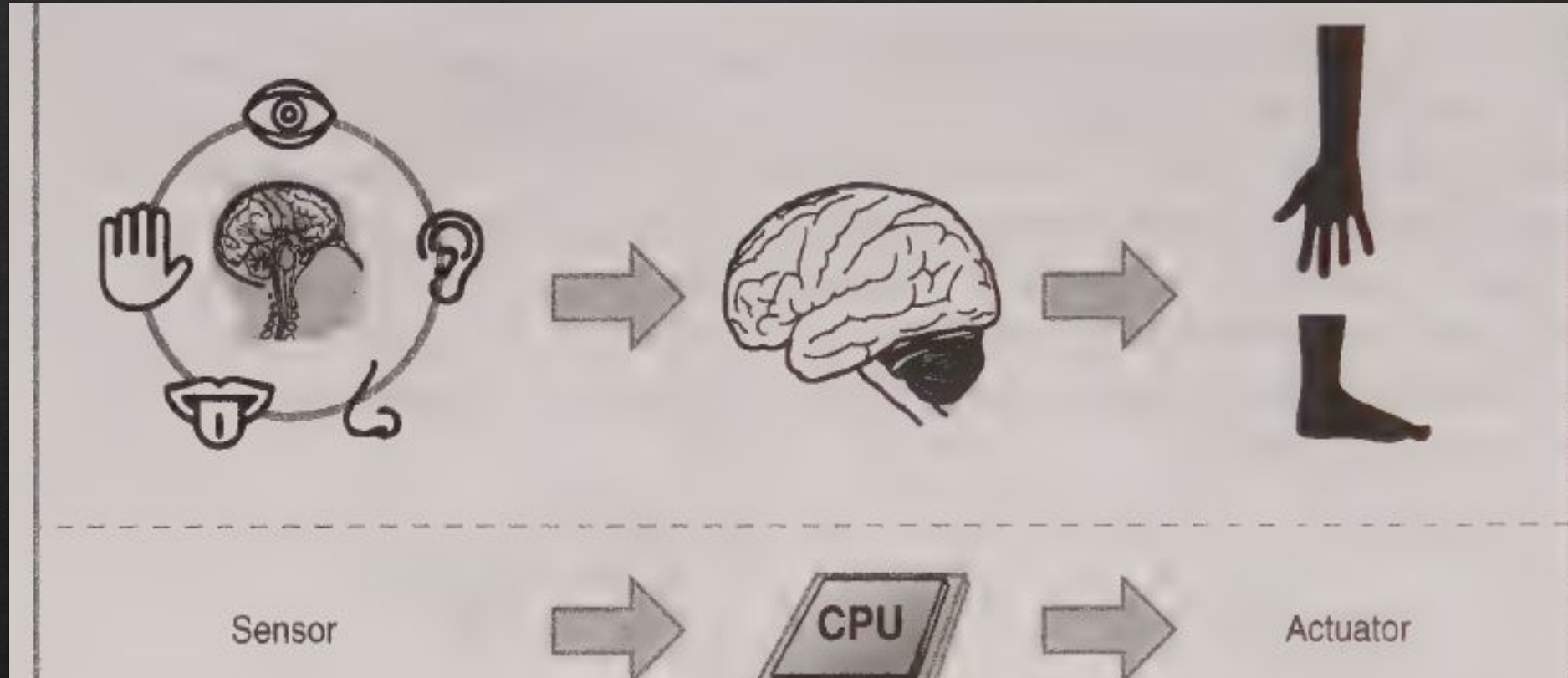
Specific Objectives	Contents
• Understand about the sensor and actuators and its types along with its characteristics	Unit 3: Sensors, Actuators, and Embedded Systems (5 hours) 3.1 Types of sensors and its characteristics (temperature, humidity, motion etc.) 3.2 Actuators (relays, motors) 3.3 Types of Microcontrollers and embedded systems in IoT (like Raspberry Pi, Arduino)

Definition of Sensor

- ❑ **Sensor** is a device that detects or measures a physical property and records, indicates, or otherwise responds to it.
 - ❑ A **sensor** is a device that measures **physical input** from its environment and converts it into data that can be **interpreted by either a human or a machine**.
 - ❑ Normally, the sensors are input devices and there are two types of sensors.
 - ❑ **Analog Sensors**
 - ❑ **Digital Sensors**
- 



Comparison of Sensor and Actuator Functionality with Humans



Sensor Types	Description	Examples
Acoustic	Acoustic sensors measure sound levels and convert that information into digital or analog data signals.	Microphone, geophone, hydrophone
Humidity	Humidity sensors detect humidity (amount of water vapor) in the air or a mass. Humidity levels can be measured in various ways: absolute humidity, relative humidity, mass ratio, and so on.	Hygrometer, humistor, soil moisture sensor
Light	Light sensors detect the presence of light (visible or invisible).	Infrared sensor, photodetector, flame detector
Radiation	Radiation sensors detect radiation in the environment. Radiation can be sensed by scintillating or ionization detection.	Geiger-Müller counter, scintillator, neutron detector
Temperature	Temperature sensors measure the amount of heat or cold that is present in a system. They can be broadly of two types: contact and non-contact. Contact temperature sensors need to be in physical contact with the object being sensed. Non-contact sensors do not need physical contact, as they measure temperature through convection and radiation.	Thermometer, calorimeter, temperature gauge
Chemical	Chemical sensors measure the concentration of chemicals in a system. When subjected to a mix of chemicals, chemical sensors are typically selective for a target type of chemical (for example, a CO ₂ sensor senses only carbon dioxide).	Breathalyzer, olfactometer, smoke detector
Biosensors	Biosensors detect various biological elements, such as organisms, tissues, cells, enzymes, antibodies, and nucleic acid.	Blood glucose biosensor, pulse oximetry, electrocardiograph

Sensors in IOT

- ❑ One of the principles of the IoT is the linking of data gathered from physical devices to the user.
- ❑ While most IoT projects are one **directional** where the device containing one or more sensors provides data for another layer in the solution but never returned to the sensor devices, it can also be a **bidirectional** flow between the user interface and the device.
- ❑ The term, transducer, is sometimes used as a synonym for sensor.
- ❑ Transducer Can be used to represent a sensor as well as an actuator
- ❑ The American National Standards Institute (ANSI) defines a transducer as “a device which provides a usable output in response to a specific measurand”.
- ❑ Jacob Fraden mentions “The term sensor should be distinguished from transducer. The latter is a converter of one type of energy into another where as the former converts any type of energy into electrical.”

Sensors and Actuators

- ❑ IOT solutions include sensors and actuators.
- ❑ An actuator is a component that converts power into mechanical energy.
- ❑ This combination of sensors and actuators can permit the solution to interact with the physical world.
- ❑ For example, it may allow the user to control hardware from a distance to interact with the physical world such as opening a door, or the manipulation of controls normally used by humans such as adding remote control to a piece of heavy equipment.
- ❑ An example of the pathway between sensors and actuators begins with an electrical signal from the sensor that is passed to a controller device where the data is amplified, conditioned and converted into a digital form, which can be processed by other nodes or devices and converted back to an electrical signal for use by the actuator

Sensor Characteristics

- Typically, IoT applications generally do not place an importance on the sensors and their characteristics, rather, it is more about how the sensor generates data and its accuracy reliability, and sample timing (speed).
- IoT applications consider the sensor a “black box” and thus, the focus is on interfacing with the sensor.
- Perhaps the most important characteristic of a sensor is its **power requirements**. Clearly, all electrical components require some form of power to work. In the case of sensors, they can either be active or passive.
- **An active sensor** typically does not generate voltage or current on its own. It requires some trigger or current to pass through to initiate a signal.
- An example of an active sensor is a thermistor or temperature sensitive resistor.
- When energized, a thermistor generates a signal of fluctuation in voltage or current based on the varying resistance of the sensor.
- It is this fluctuation in voltage that can be converted into a representation of temperature.

Sensor Characteristics

- **A passive sensor** responds to the physical environment in which it is placed without the need of an outside power source.
- One example is a piezoelectric sensor that responds to mechanical pressure resulting in small changes to its physical form that generates a charge (or change in voltage for example). Other example includes sensors that generate current from a moving a magnetic element through a coil.
- Another important characteristic of a sensor is how it physically connects to the device (such as a microcontroller or computational device and its hardware interface).
- This interface includes the physical layout of the connector (pins) as well as any additional hardware requirements such as **pull-up and pull-down resistors**, amplifiers, etc.
- Vendors supply this information via the data sheet supplied with the sensor.

Sensor Characteristics

- Another set of characteristics involves how the sensor communicates.
- **Analogue sensors** typically provide signals measured in voltage. The variance of the voltage is used to determine the range of the data (signal), which can be used to convert to a corresponding measuring unit.
- **Digital sensors** provide the data(signal) in binary form, which may be interpreted in a variety of ways depending on how the data is encoded in the binary value. The layout or protocol for encoding as well as the measuring unit is defined on the datasheet from the vendor.
- Finally, independent of the power and signal, other characteristics such as accuracy, precision, reading span, etc. also need to be considered.

How Sensor Measure

- ❑ Sensors are electronic devices that generate a voltage based on the unique properties of their chemical and mechanical construction. They don't manipulate the phenomena they're designed to measure. Rather, sensors sample some physical variable and turn it into a proportional electric signal (voltage, current, digital, and so on).
- ❑ For example, a humidity sensor measures the concentration of water (moisture) in the air. Humidity sensors react to these phenomena and generate a voltage that the microcontroller or similar device can then read and use to calculate a value on a scale. A basic, low-cost humidity sensor is the DHT-22 available from most electronic stores.
- ❑ The DHT-22 is designed to measure temperature as well as humidity. It generates a digital signal on the output(data pin). Although simple to use, it's a bit slow and should be used to track data at a reasonably slow rate(no more frequently than about once every 3 or 4s).

How Sensor Measure (example)

- When this sensor generates data, that data is transmitted as a series of high (interpreted as a 1) and low (interpreted as a 0) voltages that the microcontroller can read and use to form a value.
- In this case, the microcontroller reads a value 40 bits in length (40 pulses of high or low voltage)—that is, 5 bytes — from the sensor and places it in a program variable. The first two bytes are the value for humidity, the second two are for temperature, and the fifth byte is the checksum value to ensure an accurate read.

Analog Sensor

- **Analog sensors** are devices that generate a voltage range, typically between 0 and 5volts.
- An analog-to-digital circuit is needed to convert the voltage to a number. Most micro- controllers have this feature built in , and the Arduino is a fine example. The Arduino has a limited set of pins that operate on analog data and incorporate analog to digital (A/D)conversion circuits.
- **Analog:** Analog sensors generate an output signal or voltage, which is proportional (linearly or non-linearly) to the quantity being measured and is continuous in time and amplitude. Physical quantities such as temperature, speed, pressure, displacement, strain, and others are all continuous and categorized as analog quantities.
- For example, a thermometer or a thermocouple can be used for measuring the temperature of a liquid (e.g., in household water heaters). These sensors continuously respond to changes in the temperature of the liquid.

Digital Sensor

- ▣ **Digital sensors** like the DHT-22 are designed to produce a string of bits using serial transmission (one bit at a time). However, some digital sensors produce data via parallel transmission (one or more bytes at a time).
- ▣ These sensors generate the output of discrete time digital representation (time, or amplitude, or both) of a quantity being measured, in the form of output signals or voltages. Typically, binary output signals in the form of a logic 1 or a logic 0 for ON or OFF, respectively are associated with digital sensors.
- ▣ The generated discrete (non-continuous) values may be output as a single “bit” (serial transmission), eight of which combine to produce a single “byte” output (parallel transmission) in digital sensors.

Traditional Vs Smart Sensors

- ❑ Traditional sensors are typically used for a **single observation or event**. They tend to be designed for a specific purpose and engineered for high precision and accuracy. They may require integration with other devices to work or for power.
- ❑ Communication with traditional sensors is via a **wired (physical) connection** with a device (such as wireless transmitter or wired communication mechanism).
- ❑ Collection, aggregation, and processing of the data is typically accomplished by other nodes or devices.
- ❑ Smart sensor development has largely been driven by the needs of the IoT ecosystem.
- ❑ Sensors that combine certain other components to enable functionality such as Internet connectivity (e.g., a smart television), remote control functionality (e.g., a smart power strip and similar home automation electrical power devices), or if it has functionality to detect contextual awareness and act based on the context (e.g., a smart thermostat that can detect when humans are near and change the temperature) are considered **“smart” sensors**.

List of Sensors

- The list of sensors which is to be covered in the chapter is as follows:

MQ-02/05 Gas
Sensor



LDR



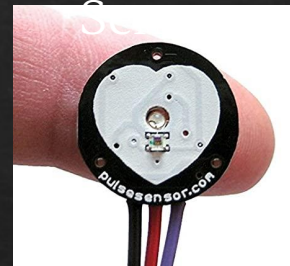
GPS



Obstacle
Sensor



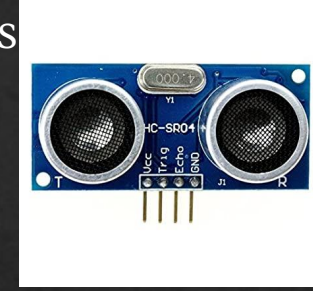
Heartbeat
Sensor



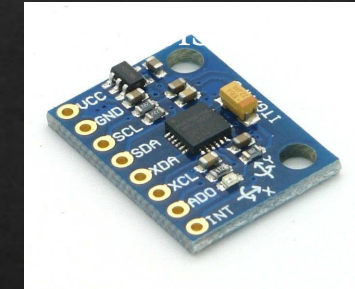
Color
Sensor



Ultrasonic Distance
Sensor



Gyro



pH
Sensor



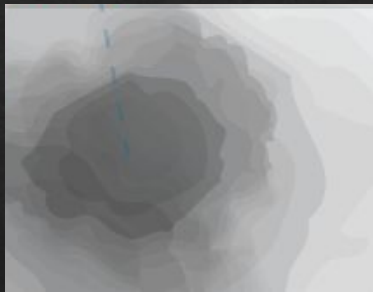
MQ-02/05 Gas Sensor

- MQ -02 sensor is used to **detect smoke**.
- H₂, LPG, CH₄ alcohol and smoke can be detected by MQ-02.
- MQ-05 is **not sensitive to smoke**, hence less used in the application.
- Pinout of MQ-02 sensor module are
 - Vcc – Connected to 5V
 - Gnd – Connected to ground
 - D0 – Digital output pin
 - A0 – Analog output pin

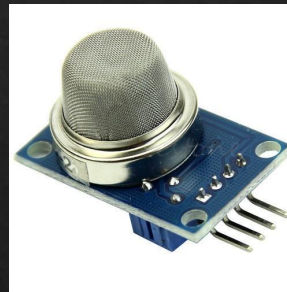
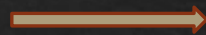


MQ-02/05 Gas Sensor – Working principle

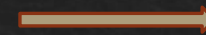
- When any flammable gas passes through the coil of the sensor, the coil burns and **internal resistance decreases**.
- This results in an **increased voltage** across it. Hence we get variable voltage at A0 pin of MQ-05 gas sensor.
- If gas present -> voltage is high.
- If gas is not present -> voltage is low.



Smoke/G
as



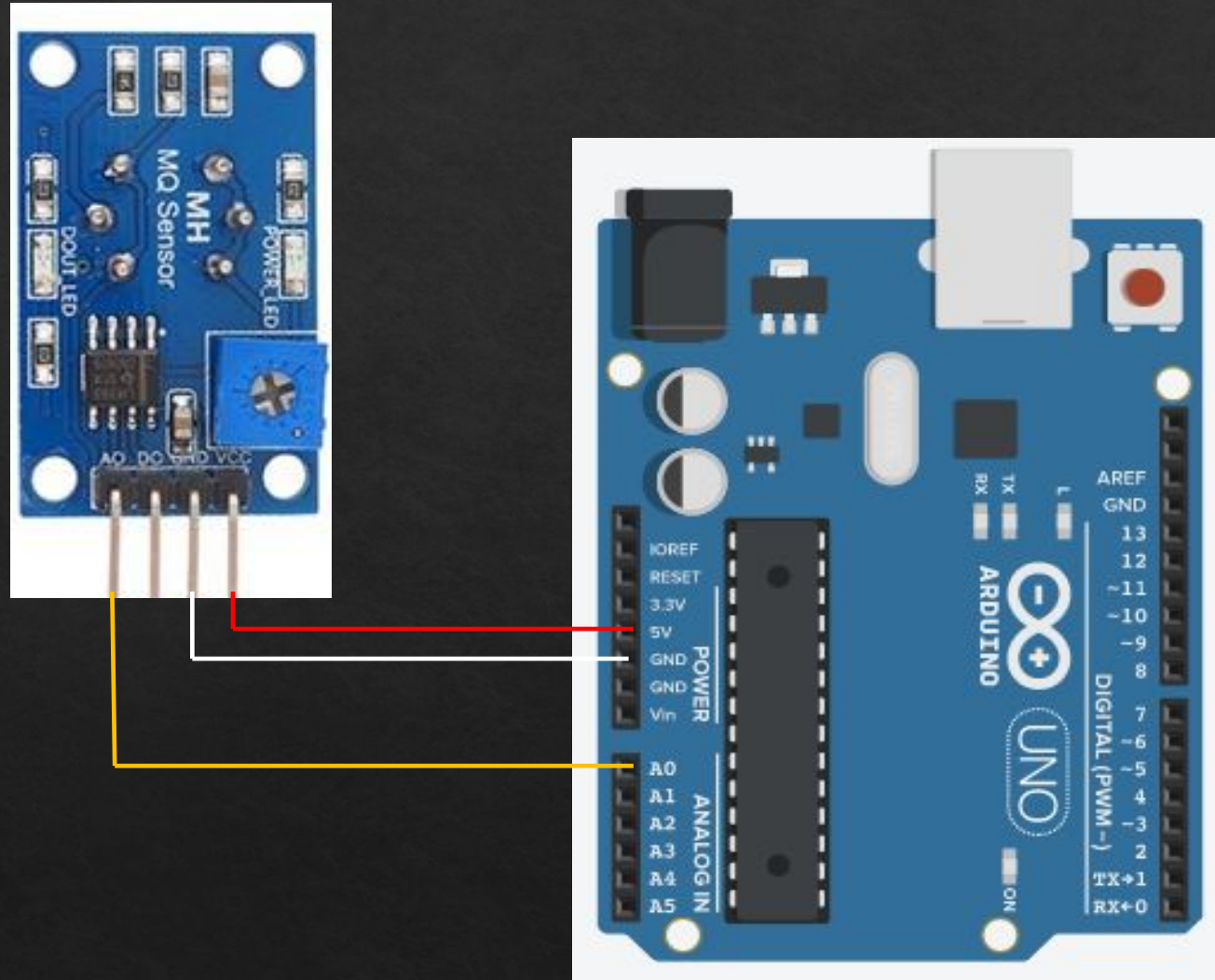
MQ-05 Gas
Sensor



Variable
Voltage

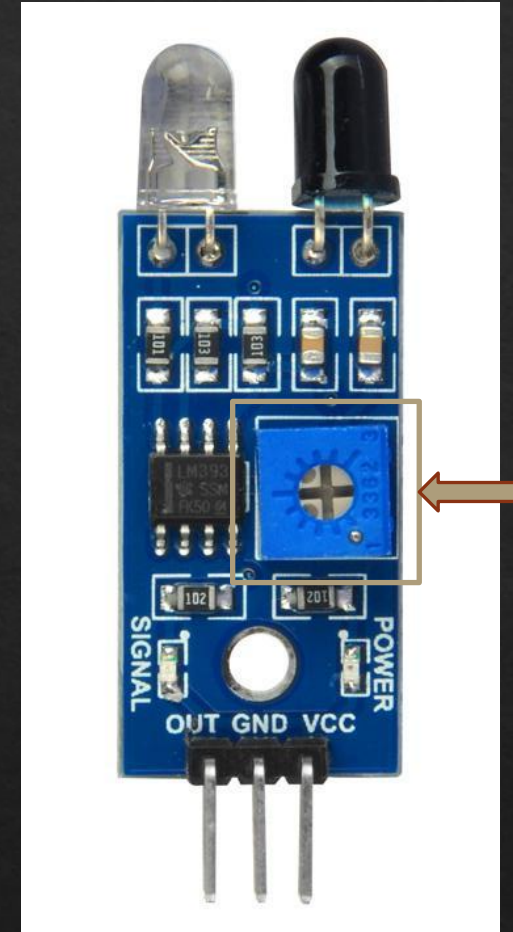
MQ-02/05 Gas Sensor – Interfacing with Arduino

- The interfacing of MQ-02 with Arduino Uno is as shown in the figure.
- Here, Vcc and Gnd pins of MQ-02 are connected to 5V and GND of Arduino board.
- We want to take the input from a Gas sensor in **an analog** format so pin A0 is connected to one of the analog pins of Arduino i.e. A0.



Obstacle Sensor

- Obstacle sensor is used for **detection of obstacle**.
- It is a digital sensor, hence gives binary output '1' or '0'.
- The **range** for detection of obstacle can be changed by **potentiometer** given.
- Pinout of obstacle sensor module are
 - Vcc – Connected to 5V
 - Gnd – Connected to ground
 - Out/D0 – Digital output pin

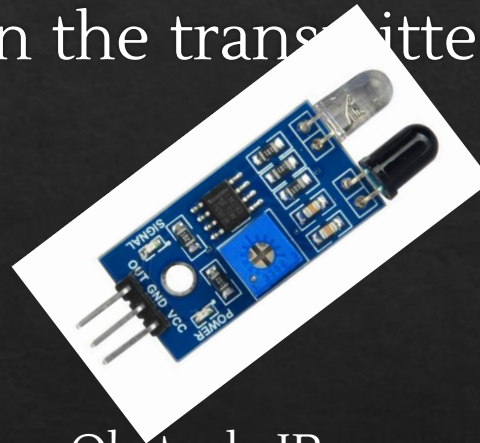


Obstacle Sensor – Working principle

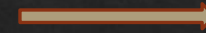
- IR emitter transmits IR signals and IR receiver receives those signals from reflection.
- If the obstacle is present then the transmitted signals are **reflected** back by obstacle and if they have amplitude greater than threshold the output signal will be '0'.
- If the obstacle is not present then the transmitted signals are **not reflected** and the output will be '1'.



Obstacle in the way



Obstacle IR Sensor



Digital Output

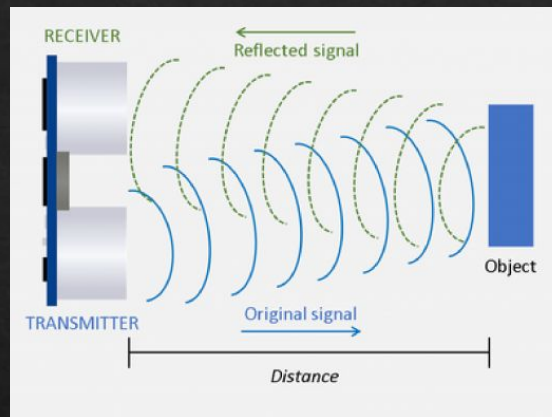
HC-SR04 Ultrasonic Sound Sensor

- Ultrasonic Sound Sensor is used to measure the **distance of an object**.
- The sensor can only measure the distance of object. It can not identify the type of object.
- It is also known as Ultrasonic distance sensor.
- Pinout of HC-SR04 module are
 - Vcc – Connected to 5V
 - Gnd – Connected to ground
 - Trigger – Generates the transmit pulse
 - Echo – Receives echo from obstacle

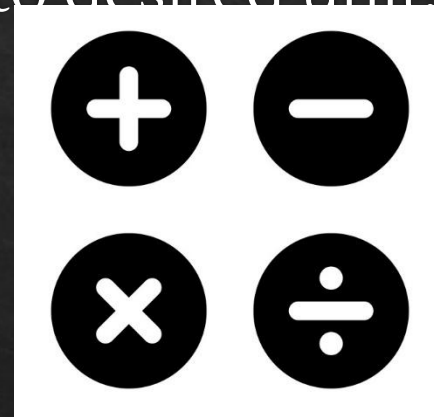


HC-SR04 Ultrasonic Sound Sensor – Working principle

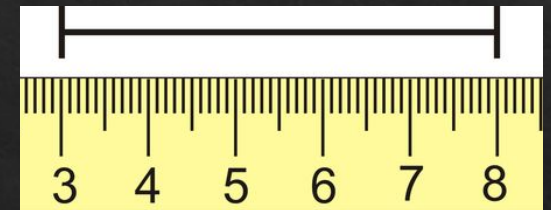
- ❑ The emitter transmits ultrasonic sound waves which are reflected by near by objects.
- ❑ The reflected pulse is received by sensor.
- ❑ Some mathematical operations are performed to obtain the distance value.
- ❑ The distance value is converted into desired unit.



Ultrasonic waves



Mathematical operations



Measurement of distance

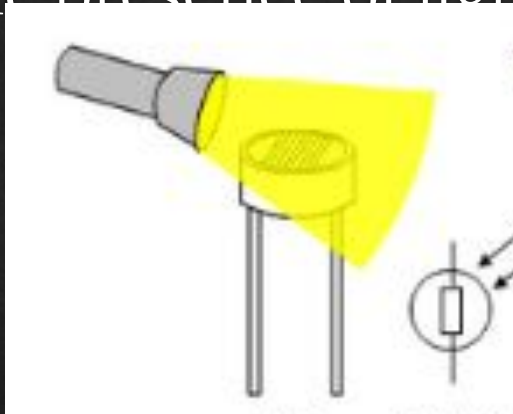
LDR Sensor

- ❑ LDR is known as Light Dependent Resistor.
- ❑ The **internal resistance** of LDR changes with respect to light intensity.
- ❑ Normally LDR works with voltage divider network.
- ❑ The LDR is used in the application where the task is performed with light intensity as input.

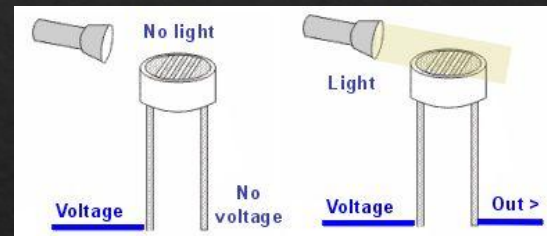
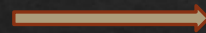


LDR Sensor – Working principle

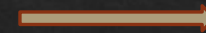
- ❑ The LDR works on principle of Ohm's law.
- ❑ As the light intensity on LDR is higher, the resistance of LDR decreases.
- ❑ The decreased resistance will drop more voltage across LDR according to Ohm's Law.
- ❑ The presence of light can be identified by the value of voltage across LDR.



Light Intensity to be measured



Output voltage measurement



Calculate the intensity

Heartbeat Sensor

- ▣ Heartbeat sensor is having typical application in health care domain.
- ▣ This is mainly used in wearable devices to monitor the heart beat / heart rate of the person.
- ▣ The sensor is inexpensive and generates square waves for each pulse. The averaging of pulses gives analog voltage according to the heartbeat.
- ▣ Pinout of heartbeat sensor module are
 - ▣ Vcc – Connected to 5V
 - ▣ Gnd – Connected to ground
 - ▣ A0 – Analog output pin

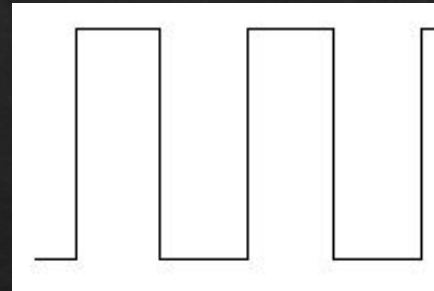
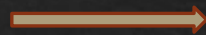


Heartbeat Sensor – Working principle

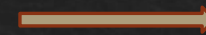
- ❑ The sensor senses the heartbeat as per blood circulation in the body.
- ❑ It generates the output pulse signal according to the rate at which heart beats.
- ❑ The output pulses are considered as PWM waves. Hence they are converted into analog signals.
- ❑ This analog signal is converted to digital which gives the output beat rate.



Heartbeat
input



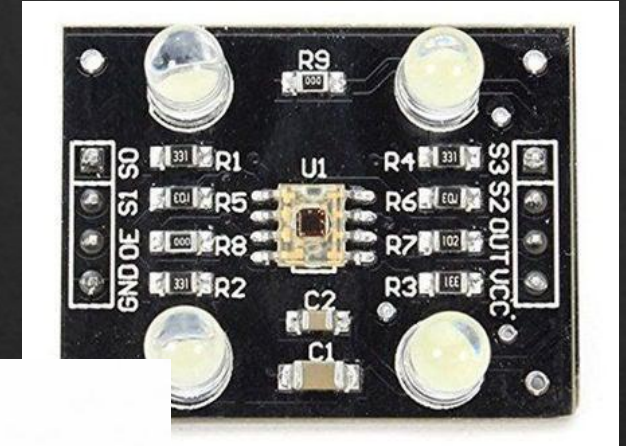
Generation of Digital
Pulse
as output



Calculates the
BPM

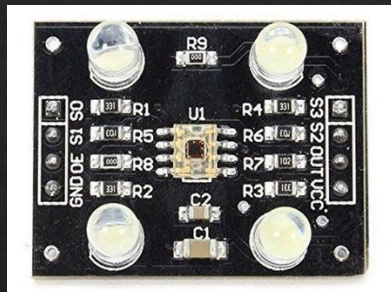
Colour Sensor

- Colour sensor is used to detect the RGB colour coordinates of a particular colour.
- The colour sensor module has TSC3200 IC that converts colour to frequency by enabling a particular colour diode turn by turn.
- The colour sensor is mainly used in the application where the objects are identified by their colour.
- Pinout of colour sensor module are
 - Vcc – Connected to 5V
 - Gnd – Connected to ground
 - S0, S1 – Used for output frequency scaling
 - S2, S3 – Type of photodiode selected

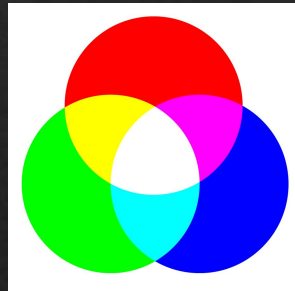
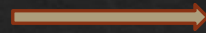


Colour Sensor – Working principle

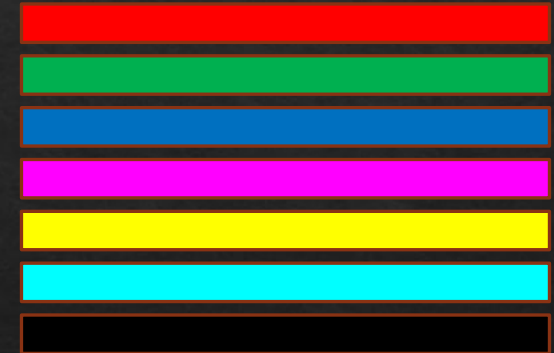
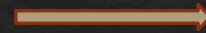
- ❑ The sensor works by imparting a bright light on an object and recording the reflected colour by the object.
- ❑ The selected photodiode for colour of red, green and blue converts the amount of light to current.
- ❑ These RGB values are further processed to identify the exact colour combination.



Input to
photodiode
of colour sensor



Output RGB colour
values



Calculation of
actual
colour of object

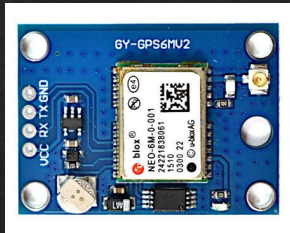
GPS Sensor

- ❑ GPS sensor is used to get the location data of the place where sensor is situated.
- ❑ The data from sensor is acquired in National Marine Electronics Association (NEMA) format. The format is used by marine department for communication.
- ❑ There are multiple variants of GPS sensors available in the market. We can select based on our requirements.
- ❑ Here, the given sensor is GY-GPS6MV2 as shown in the image.
- ❑ Pinout of GPS sensor module are
 - ❑ Vcc – Connected to 5V
 - ❑ Gnd – Connected to ground

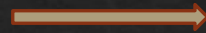


GPS Sensor – Working principle

- ❑ The sensor is placed in the system whose location is to be tracked.
- ❑ The GPS device communicates with GPS satellite and the satellite returns its current location in form of latitude and longitude.
- ❑ The sensor transmits the data to the microcontroller in a predefined format from which we can extract the tracked location.

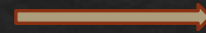


GPS device with
antenna
for communication



Latitude
Longitude

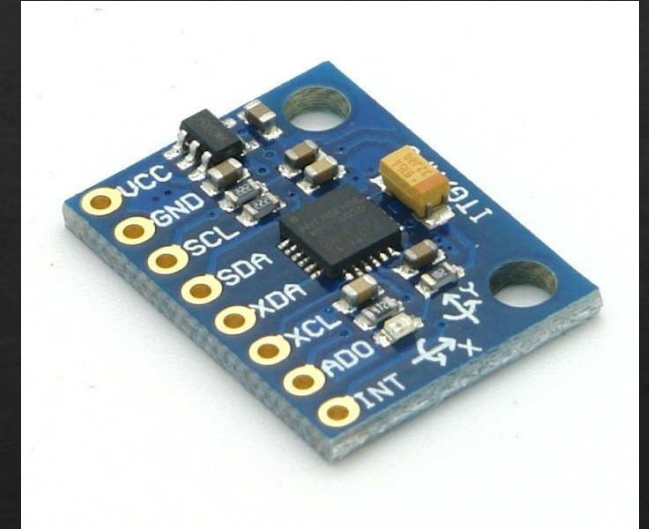
Returns latitude and
longitude
as output



Tracked
Location

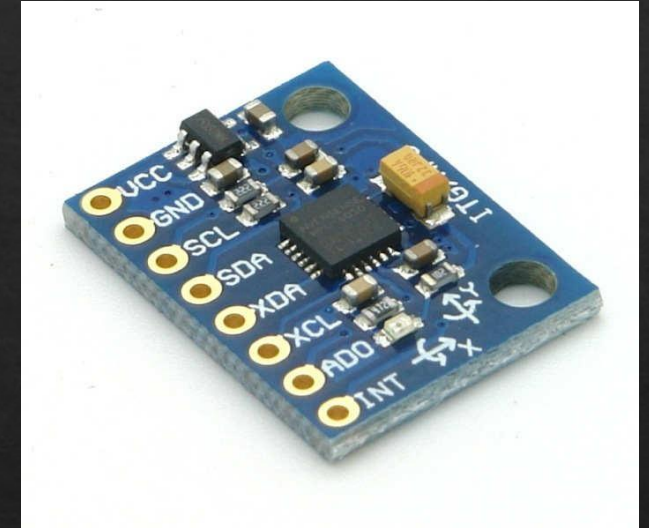
Gyro Sensor

- ❑ The Gyro sensor is very useful in **wearable devices**.
- ❑ It is used to find the position or angle of rotation of the body.
- ❑ It mainly senses
 - ❑ Rotational motion
 - ❑ Changes in orientation
- ❑ The Gyro sensor communicates with microcontroller with I2C (Inter-Integrated Communication).
- ❑ I2C is the most sophisticated two-wire communication protocol.
- ❑ It works with the device address and its respective data frames.



Gyro Sensor (Cont.)

- The pin out of Gyro sensor MPU6050 are as following.
 - Vcc – connected to 5V
 - GND – connected to ground
 - SCL – Serial Clock Line
 - SDA – Serial Data Line
 - XDA – Auxiliary Serial Data
 - XCL – Auxiliary Serial Clock
 - AD0 – This pin can be used to vary the address incase of multiple sensors used
 - INT – Interrupt pin available for certain application



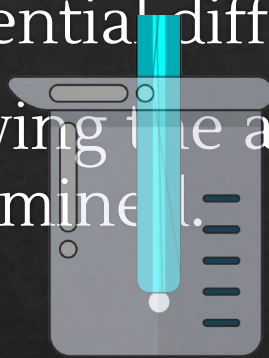
PH Sensor

- ❑ The pH sensor is used to detect hydrogen ions concentration of a liquid.
- ❑ pH sensor are mostly used in laboratories to test the acidity of solution.
- ❑ By the use of pH sensor we can identify the pH of a solution and whether it is acid or base.



pH Sensor – Working principle

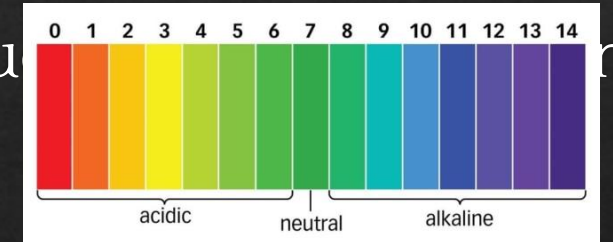
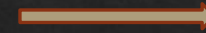
- When this sensor is placed in a solution, the smaller ions penetrate the boundary area of glass and the larger ions remain in the solution. This creates potential difference.
- The pH meter measures the difference in electrical potential between the pH electrodes.
- The potential difference generates different analog values for different liquids.
- By knowing the analog value of standard water, the pH value can be determined.



pH Sensor
with
Electrodes



Output
Voltage



Determination of
alkalinity
of the liquid

Actuators

- ▣ Actuators are devices that convert an electrical power source to mechanical energy in the form of controlled physical motion.
- ▣ Examples of actuators include motors that open and close doors or windows, open and close valves in gas and liquid fixtures, and other mechanical movements that can manipulate switches and dials in machinery or appliances.
- ▣ Recent advances in actuator designs have given rise to actuators that can convert mechanical movement into energy.
- ▣ These types of actuators are being used in use cases where mechanical motion can be used to generate power for storage in battery packs thereby reducing the energy footprint of the solution.

Actuators

- ▣ Actuators can be combined with sensors to enable developers to cover the gap from the digital world to the physical. For example, solutions that encompass remote access to machinery that allow operators from a different location to interact with machinery and hardware in another location.
- ▣ Sensors can measure an event converting mechanical energy to electrical energy, which can then be converted to a digital signal, which can then be processed and some action determined that can, in turn, require converting a new digital signal to electrical for use in a different actuator to complete the action required by the original sensor value.

Actuator Types

1) **Hydraulic** : A hydraulic actuator works on the principle of compression and decompression

of fluids. These actuators facilitate mechanical tasks such as lifting loads through the use of hydraulic power derived from fluids in cylinders or fluid motors.

2) **Pneumatic** : A pneumatic actuator works on the principle of compression and decompression

of gases. These actuators use a vacuum or compressed air at high pressure and convert it into either linear or rotary motion.

3) **Electrical** : electric motors are used to power an electric actuator by generating mechanical torque.

4) **Thermal/magnetic** : The use of thermal or magnetic energy is used for powering this class of actuators.

5) **Mechanical** : In mechanical actuation, the rotary motion of the actuator is converted into linear motion to execute some movement.

Popular actuators used for IoT-based control applications



(a) Brushless DC motor (b) Brushless DC motor (c) Stepper motor (d) Geared stepper motor (e) DC motor



(f) Relay array (g) Hydroelectric generator (h) Hydroelectric generator (i) Solenoid-based flow valve (j) Solenoid-based flow valve

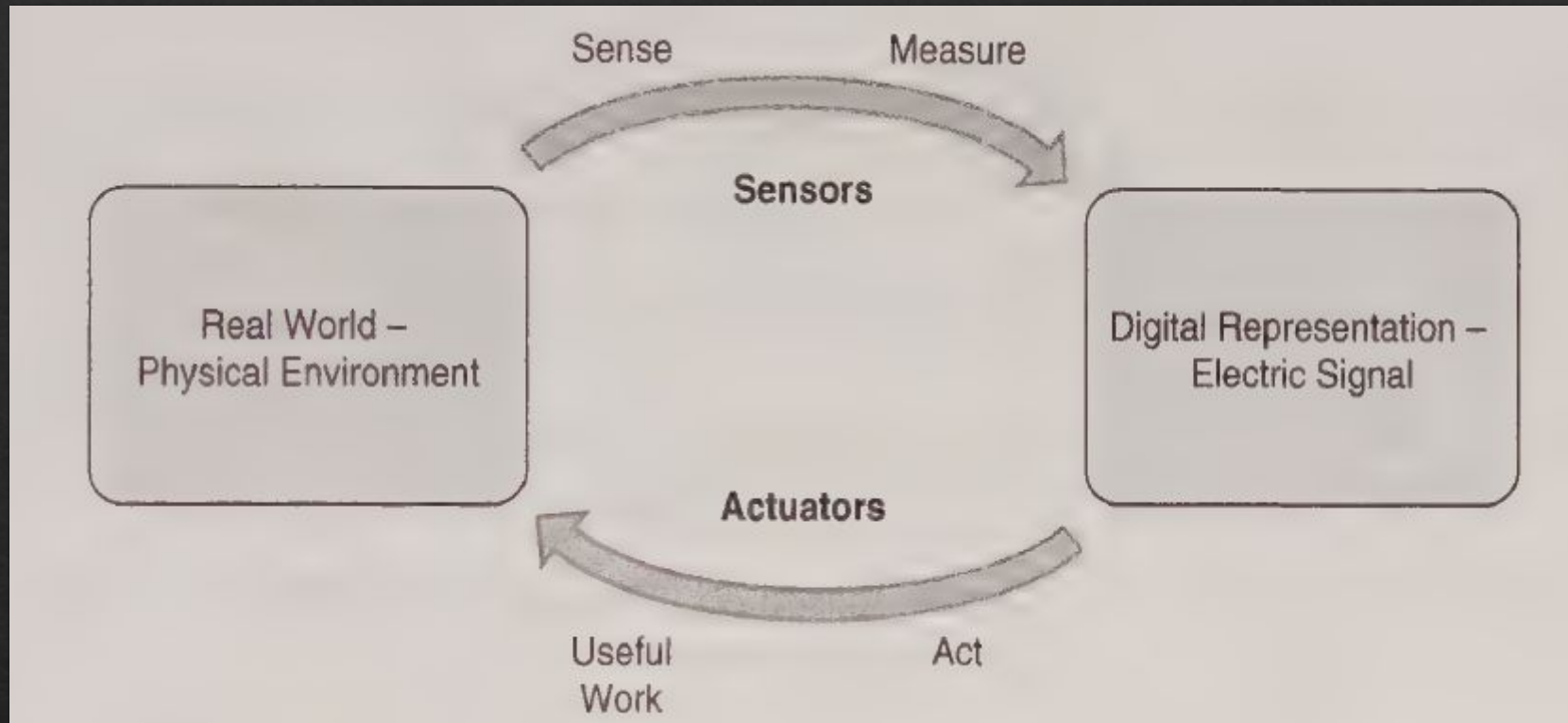


(k) DPDT switch (l) Push button switch

Actuators and Sensors

- ▣ With the proliferation of actuators and sensors, system faults have become a design and operational priority.
- ▣ In industrial systems, undetected faults in the actuator or sensor can cause safety concerns as well as product control and quality concerns whereas faults that generate a false alarm (or false failure event) can impact productivity.
- ▣ As actuators and sensors become more advanced with digital interconnectivity such as wireless communication and advances in hardware in the areas of miniaturization and energy consumption, actuators are evolving into smart devices that permit real-time monitoring of state and operation.
- ▣ When combined, actuators and sensors can form a smart actuator making it easier to use and integrate into solutions. Together with the real-time monitoring capabilities, smart sensors present another field of advanced research.

How Sensors and Actuators Interact with the Physical World



- sensors are designed to sense and measure practically any measurable variable in the physical world. They convert their measurements (typically analog) into electric signals or digital representations that can be consumed by an intelligent agent (a device or a human).
- Actuators, on the other hand, receive some type of control signal (commonly an electric signal or digital command) that triggers a physical effect, usually some type of motion, force, and so on.

Smart Vs Traditional Actuators

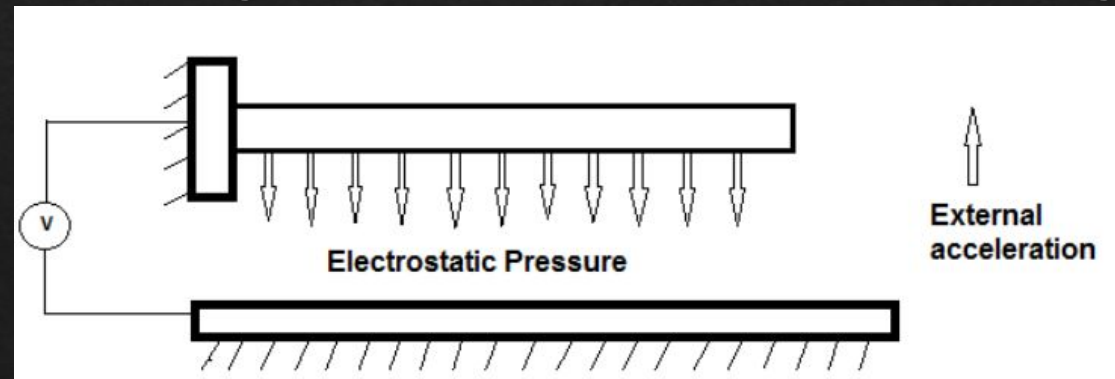
- Since their adoption in modern technologies, actuators have been used to extend human capabilities. Actuators enable humans to do more delicate work, create robots to execute repetitive tasks in manufacturing such as pick-and-place machines, and do more beyond the limits of human strength and endurance by operating heavy equipment such as agricultural machines.
- Smart Actuators are actuators combined with sensing capabilities combining to form a new category of actuators that enable actuators to perform predictive motion based on the sensory input. For example, they can act like muscles in an arm or leg to move the limb a certain way when the sensory computational result dictates such a movement .
- The combination of these technologies permits devices that can be self-contained and enable more operations to be managed more easily from a common source such as an intermediate computational node, an external source, or a programmable logic controller (PLC), or a dedicated control unit.

Characteristics of Actuators

- ▣ There are four characteristics of actuators based on operating principles. These include electrostatic, piezoelectric, thermal, and magnetic.
- ▣ The stroke or length of the actuator movement can range from sub-nanometer to centimeter scales where each increment is the minimal distance for the actuator.
- ▣ These categories represent those actuations used in mechanism from the field of micro-or nano-position requirements.

Electrostatic Actuators

- The most common class of actuators are the class of electrostatic actuators. There are many designs that use a similar pair of conducting electrodes arranged so that if mutual capacitance occurs, an attractive force is generated when voltage is applied between the electrodes.
- Electrostatic actuators have a force output between 10^{-6}N .. 10^{-3}N , but typically can achieve large displacements up to $200\text{ }\mu\text{m}$.
- Advantages are low power consumption, fast response
- Disadvantage is they typically require larger voltages to operate, perhaps in the range of several hundred volts making them more suited to high-voltage electronics. Small displacement



Piezoelectric Actuators

- A class of actuators that provide higher precision are Piezoelectric actuators.
- **Direct Piezoelectric Effect:** Generate an **electrical charge** when subjected to **mechanical stress or pressure**.
- **Converse Piezoelectric Effect:** Undergo **mechanical deformation (change in size or shape)** when an **electrical field** is applied across them. This is the principle behind piezoelectric actuators.
- Their maximum force is typically between $10^{-5}\text{N} \dots 10^{-3}\text{N}$ and a displacement of $10^{-7}\text{m} \dots 10^{-3}\text{m}$. They are typically built by stacking piezoelectric disks plates that permit small displacement with high force when voltage is supplied.
- There are many different designs of piezoelectric actuators each with characteristics specific to precision position.
- Advantages are high precision, fast response, generation of high force
- Disadvantages are small displacements, High voltage requirement

Thermal Actuators

- ▣ Actuators that react to changes in temperature either in their materials or through motion are called thermal actuators.
- ▣ They are typically used where a high force output under low voltages and can deliver a broad range of displacement and force such as those thermal actuators that use polymers in their construction that maximizes thermal expansion.
- ▣ Thermal actuators that operate at nanoscale are used with nanoscale structures where motion is generated by expansion and contraction of the materials.
- ▣ Advantages are High force generation, simple design and large displacement
- ▣ Disadvantages are slow response, high power consumption, temperature sensitivity

Magnetic Actuators

- ❑ Magnetic actuators are in a class of linear, rotary, or oscillatory motion mechanisms. They use moving iron coil, or magnet mechanisms for motion.
- ❑ The disadvantage of magnetic actuators is that they do not scale into the microscale.
- ❑ Furthermore, the force achieved is within the range of 10^{-7}N ... 10^{-4}N maximum, while the displacement achieved is within the range of 10^{-5}m and 10^{-3}m maximum.
- ❑ Advantages are large displacement, High force and fast response
- ❑ Disadvantages are high power consumption, Bigger size and Electromagnetic Interference (EMI)

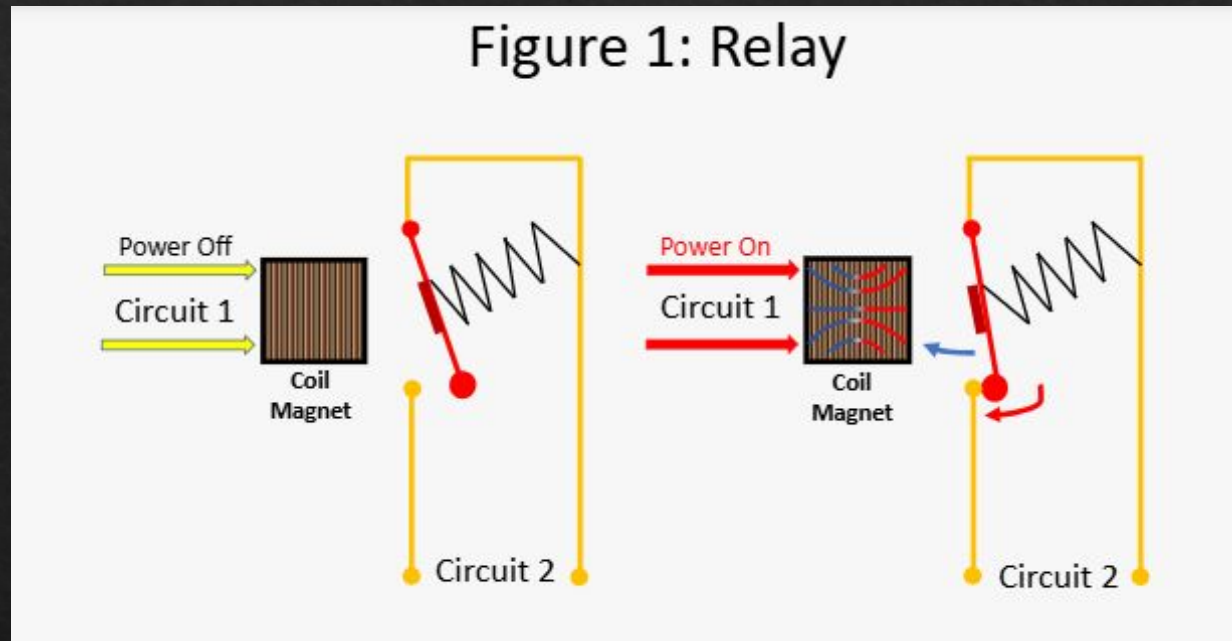
Wireless Sensor and Actuator Networks (WSANs)

- ▣ Electrical actuation has been the evolution from mechanical actuation starting 50 years ago. The primary goal has been to improve data collection about mechanical operations by using external, electrical components to monitor movement and position of devices. In fact, this has resulted in the incorporation of electronic circuits (often integrated circuits) into the actuators themselves. This has allowed the incorporation of switching, positional feedback, and diagnostics into the actuators. Adding wireless communication functionality to sensors as well as actuators has made them easier to integration in IoT solutions.
- ▣ WSANs allow for the distribution of sensors and actuators through wireless connections. This permits a decoupling of the solution components that increases sophistication, simplifies operation, and lowers the cost of ownership. This permits the creation of remote, distributed interactions with the physical world.
- ▣ WSANs are more than a complement or enhancement for wireless sensor networks (WSN).

Actuators - Relay

Relay: A relay is an electromechanical actuator that functions as a switch, controlling high-power devices using a low-power signal.

- It uses an electromagnet coil to mechanically open or close electrical contacts, allowing an IoT system to control devices like lights, fans, or motors



Relay

Advantages

- ❑ High Power Handling: Can control devices up to 250V AC, 10A, suitable for appliances
- ❑ Isolation: Provides electrical isolation between the control circuit (e.g., 5V) and the load circuit (e.g., 220V), enhancing safety.
- ❑ Cost-effective

Disadvantages

- ❑ Slow Response- Electromechanical relays have a switching time of 5-15 ms, which may be too slow for real-time IoT applications
- ❑ Mechanical Wear
- ❑ Power Consumption-The coil requires continuous power to stay on

Actuators-Motor

Motor: Motors are actuators that convert electrical energy into mechanical motion.

- ▣ Motors operate based on electromagnetic principles, where a current-carrying conductor in a magnetic field experiences a force (Lorentz force), producing motion.
- ▣ In IoT, motors enable physical movement, such as opening a smart lock, rotating a camera, or driving a robotic arm.

Types of Motors in used in IoT

- ▣ DC Motors: Simple motors that rotate continuously when a DC voltage is applied.
 - Used in IoT-based robotic cars for wheel movement
- ▣ Servo motors: sophisticated closed-loop systems that provide precise control over angular position, velocity, and acceleration.
 - In a smart security camera, a servo motor rotates the camera to a specific angle based on motion sensor data

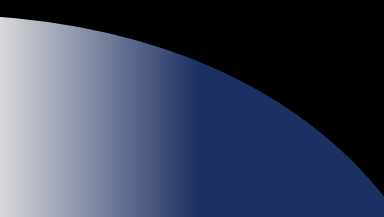
Motors

Advantages

- ▣ Versatility: Motors support a wide range of motions (rotation, linear, precise positioning), fitting diverse IoT applications.
- ▣ Precise Control: Servo motors offer high precision, critical for IoT systems like smart cameras or robotics
- ▣ Scalability: used for high power-requirements to low costs projects

Disadvantages

- ▣ Power consumption: Motors consume significant power, draining batteries in IOT devices
- ▣ Complexity
- ▣ Heat and Noise



Embedded System

Embedded System

- ◆ An Embedded System is a computer system that has computer hardware and software embedded to perform specific tasks.
- ◆ In contrast to general purpose computers or personal computers (PCs) which can perform various types of tasks, embedded systems are designed to perform a specific set of tasks.
- ◆ Key components of an embedded system include,
 - Microprocessor or Microcontroller, memory (RAM, ROM, cache), networking units (Ethernet, Wi-Fi adapters), input/output units (display, keyboard, etc.) and storage (such as flash memory). Some embedded systems have specialized processors such as digital signal processors (DSPs), graphics processors and application specific processors. Embedded systems run embedded operating systems such as real-time operating systems (RTOS).

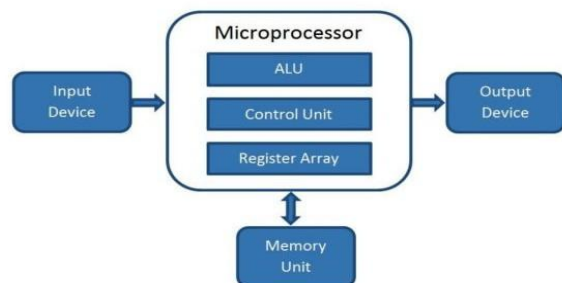
MICROPROCESSOR (MPU)



MICROCONTROLLER (MCU)



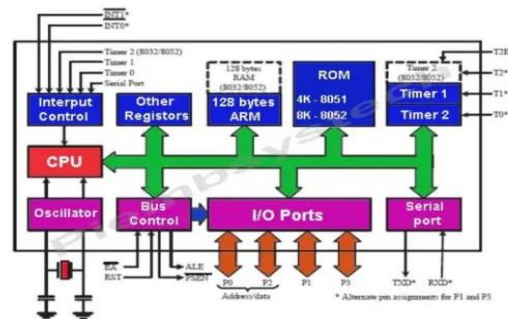
- ✓ Brain of general-purpose computing devices
- ✓ Needs external memory & peripherals
- ✓ Used in PCs, servers, and complex applications
- ✓ Higher cost (due to need for external components)
- ✓ Higher power Consumption
- ✓ Higher clock speed, more processing power
- ✓ External RAM and ROM required
- ✓ More complex due to separate components
- ✓ Intel Core i7, AMD Ryzen, ARM Cortex-A



- ✓ **Microprocessor** is like a powerful brain in a desktop computer—great at heavy thinking but needs external help for basic actions.



- ✓ Compact all-in-one chip with CPU, memory & I/O
- ✓ Perfect for real-time embedded applications
- ✓ Commonly used in smart devices, automotive ECUs, home automation, etc.
- ✓ Lower cost (everything integrated)
- ✓ Lower Power Consumption
- ✓ Optimized speed for control-based applications
- ✓ Internal memory (RAM, ROM/Flash)
- ✓ Simpler design and easy to implement
- ✓ ATmega328 (Arduino), STM32, PIC, 8051

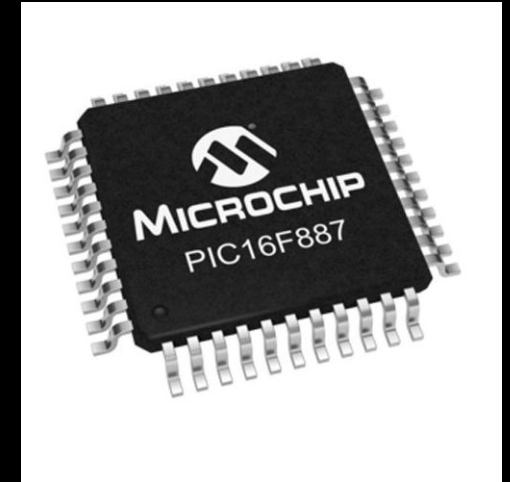
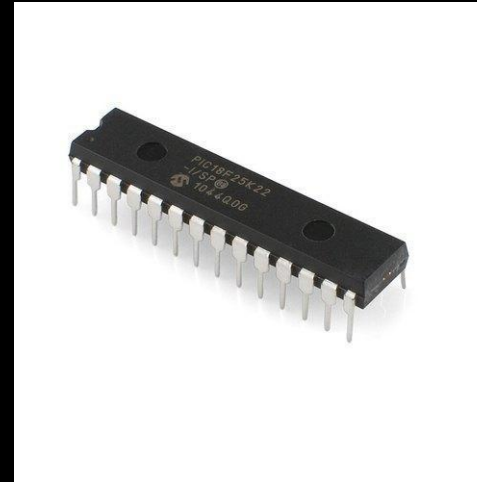


- ✓ **Microcontroller** is like a self-sufficient brain in a washing machine—it controls, decides, and executes tasks all on its own.

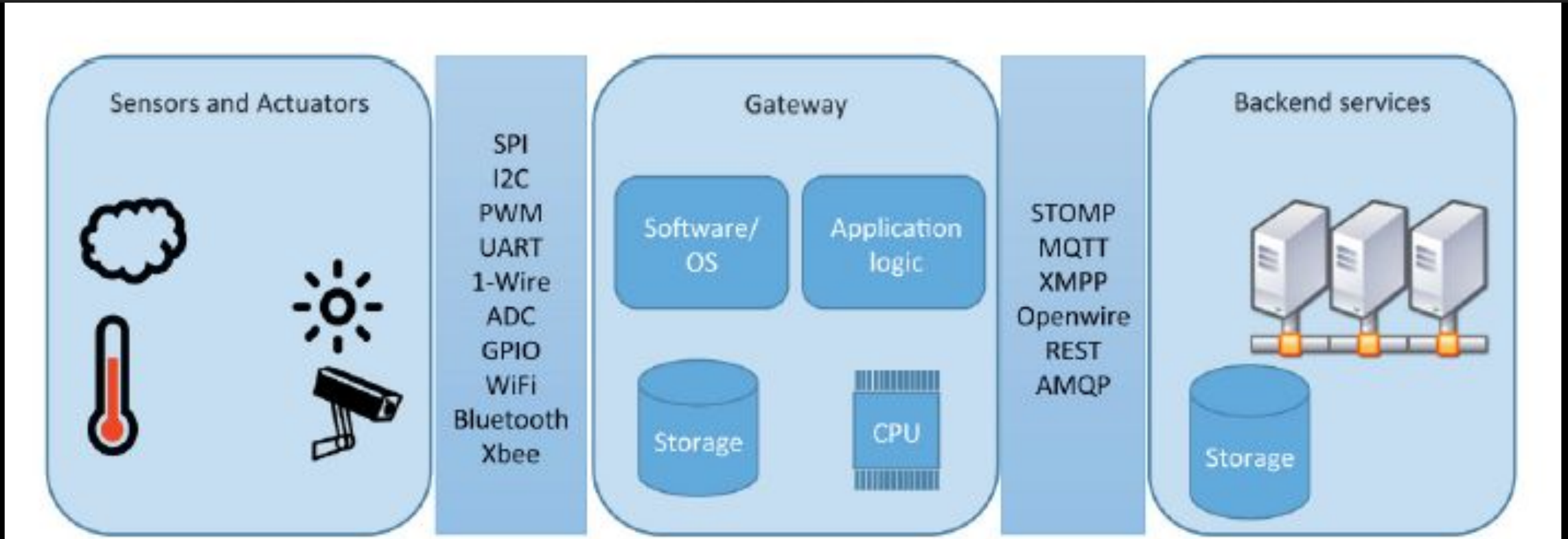
Overview of Microcontrollers

- ▣ **Microcontroller** is a digital device consisting of CPU, peripheral I/Os, memories, interrupts, etc. in a single chip or IC.
- ▣ Microcontroller is a **compact integrated circuit** designed for a specific operation in an embedded system.
- ▣ The examples of various microcontrollers are given as:

Atmel AVR	Microchip – PIC	Intel 8051
AtMega8	PIC18F452	AT89C51
AtMega328P	PIC16F877	AT89S52
AtMega16	PIC16F676	AT89c2051
AtMega2560	PIC16F72	P89V51



Applied IoT



- ◆ The gateway is responsible for communicating with the sensors and actuators as well as the backend services; it is a translator between localized interfaces, such as hard-wired sensors and remote backend systems.

Applied IoT

- ❖ The gateway is optional in IoT sensor-devices that can communicate directly with the Internet; either the sensor becomes very lightweight and has minimal functionality, or some of the desired gateway functions such as data persistence and security are incorporated into the sensor.
- ❖ The backend services are predominantly used to store the IoT device data but can include additional functionality such as individual device configuration and analysis algorithms. It is beneficial to also consider data analysis as part of the backend, since the data is only useful if it can be analyzed.

GATEWAY HARDWARE

- ❖ **Microprocessor** (MPU) usually only offers processing power and requires RAM and storage to be added as separate chips on a PCB.
- ❖ This results in a lot of additional work and power, whereas a **microcontroller (MCU)** has most of the basics embedded on a single chip; it is for this reason that we consider microcontrollers for the gateway hardware.
- ❖ More powerful microcontrollers exist, and are often referred to as a System on a Chip (SoC).
- ❖ Both a microcontroller and a SoC are of little use on their own; they need to be assembled on a PCB with other components, for example, power regulators, pin headers, and peripheral interfaces. The easiest way to build a prototype is to use a development board. The development board, often called an evaluation board, will expose all the electronic interfaces, and provide a way to program and power the chosen chip.



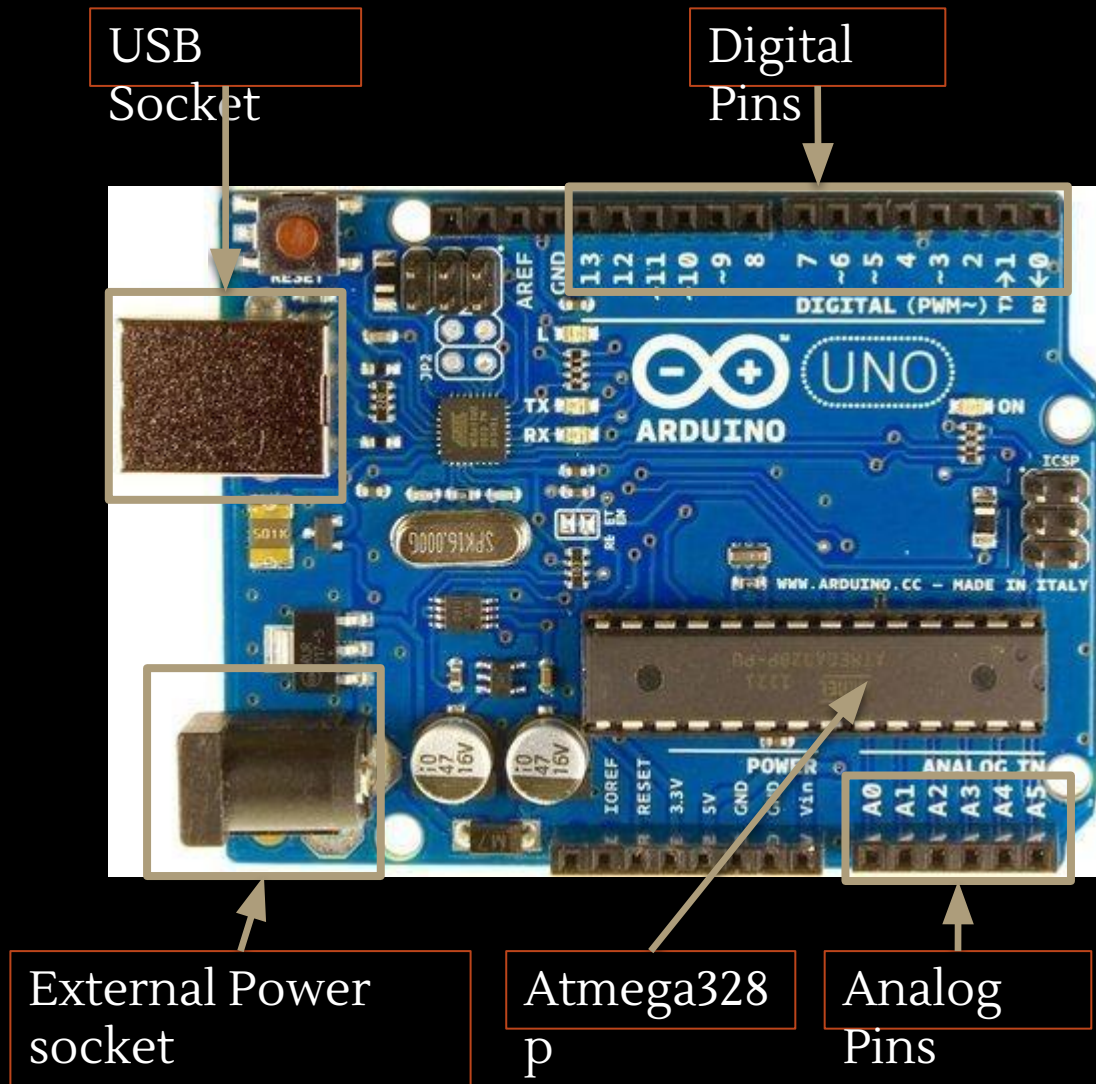
Arduino UNO Development Board With ATmega328 Microcontroller

Arduino

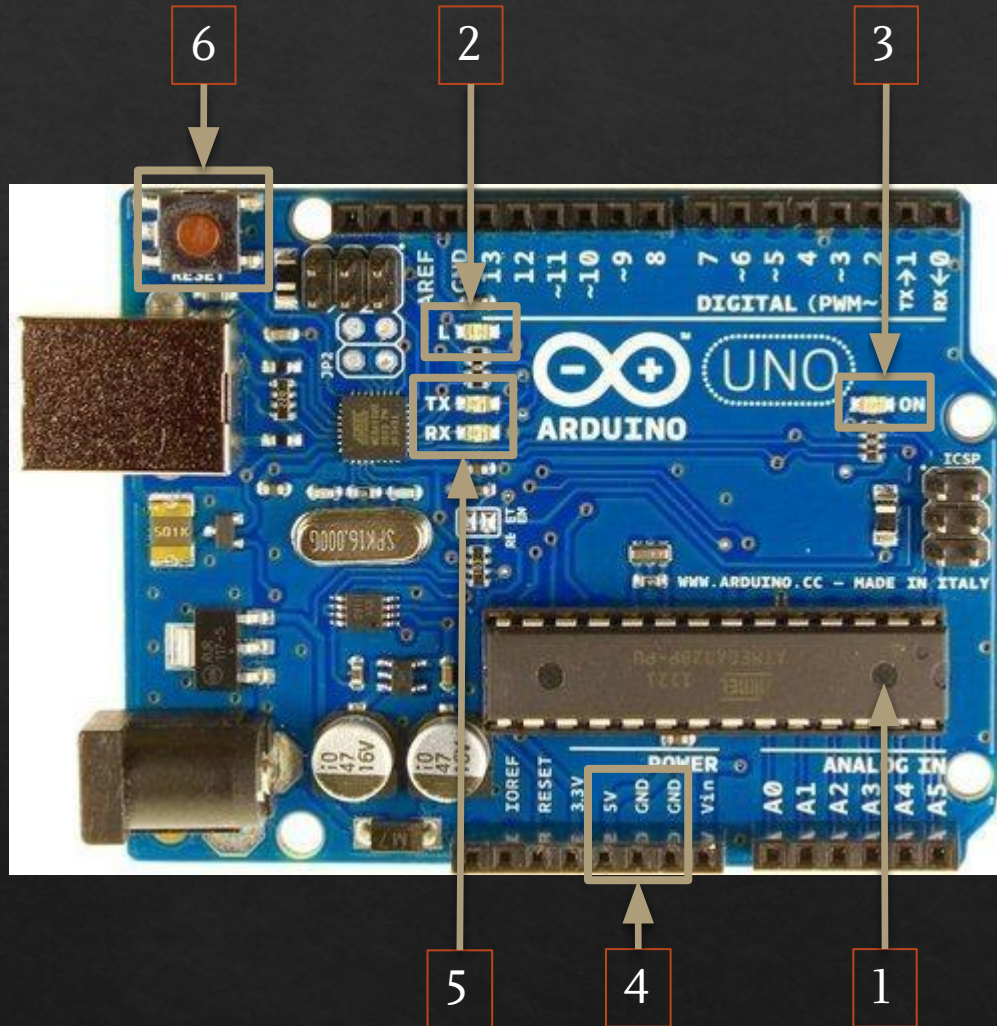
- ❖ Arduino is an open-source platform which comprises both hardware and software modules. The hardware are typically processor boards with analog and digital pins for various functions.
- ❖ Arduino IDE (integrated development environment) is the software that is used to communicate to the hardware and program them to perform specific functions.

Arduino Uno

- Atmega328p Microcontroller inside
- The operating voltage is 5V which is given by
 - USB socket or
 - External power socket
- DC Current for each input/output pin is 40 mA
- Digital input/output pins are 14 to interface digital input or output devices
- There are 6 Analog i/p pins used for interfacing analog input devices.
- What is Digital and Analog?
- Memory Components:
 - 32KB flash (program) memory
 - SRAM is 2 KB
 - EEPROM is 1 KB



Components in Arduino Uno



1. Atmega328p Microcontroller: The heart of the board.
2. On-board LED interfaced with pin 13.
3. Power LED: Indicates the status of Arduino whether it is powered or not.
4. GND and 5V pins: Used for providing +5V and ground to other components of circuits.
5. TX and RX LEDs: These LEDs indicate communication between Arduino and computer.
6. Reset button: Use to reset the ATmega328P microcontroller

Raspberry Pi

- ❖ The Raspberry Pi is a single-board personalized computer developed by the Raspberry Pi Foundation (a charitable organization) in the UK in the year 2009.
- ❖ The main motive of the developers was to promote the study of computer science and its exposure to computers among people of all ages at a low cost.
- ❖ The Raspberry Pi is comparable to the size of a credit card and equipped with an HDMI (high-definition multimedia interface) port for connecting with regular monitors/TVs. It attaches to a standard keyboard and mouse through USB ports. It also has an Ethernet port, is Wi-Fi enabled, and also has a Bluetooth.
- ❖ The Raspberry Pi also features 40 general-purpose input/output (GPIO) pins, which is usable for a wide range of applications, from lighting an LED bulb to building an HD media center to creating a gaming machine.

Raspberry Pi Architecture

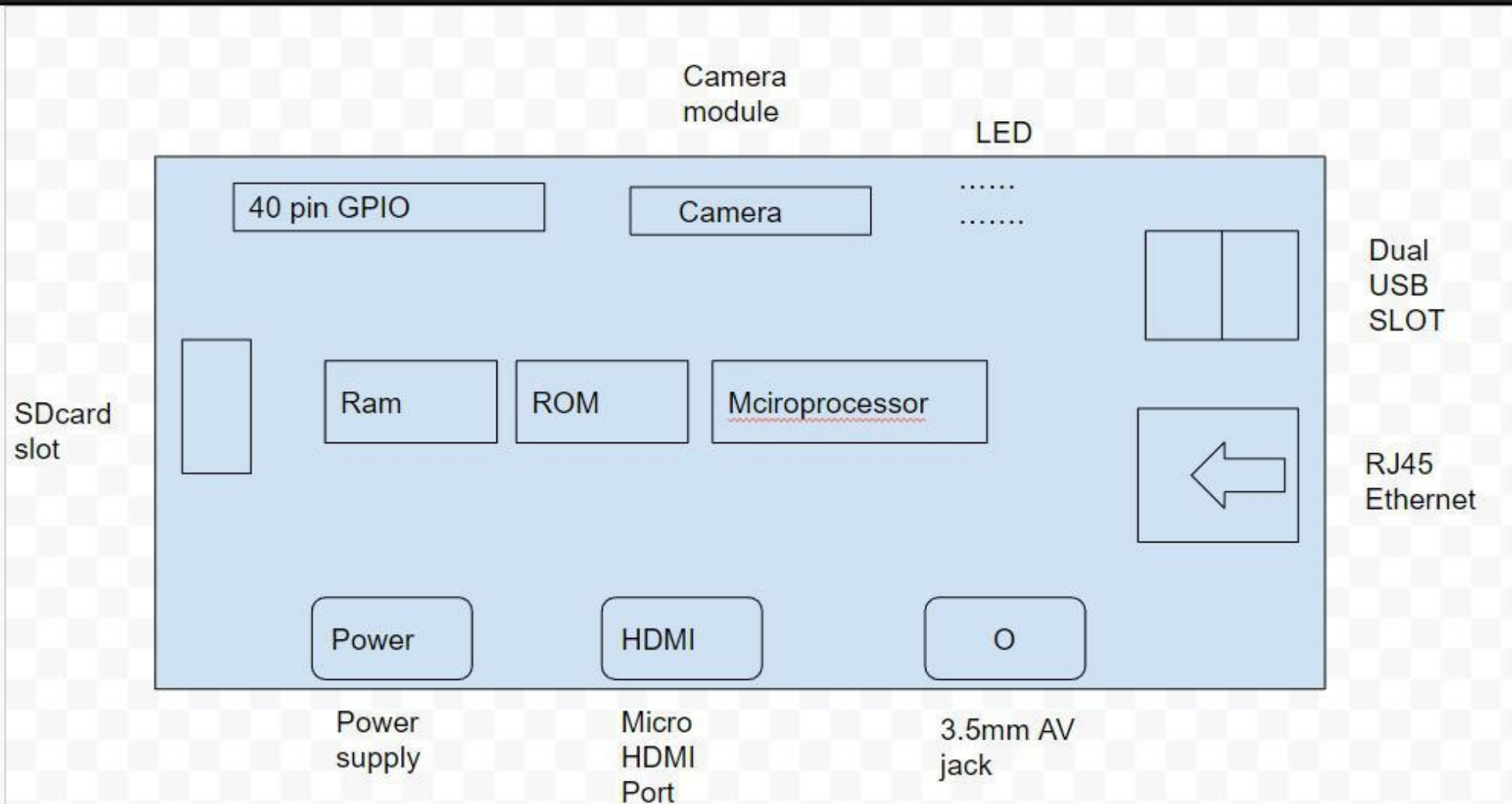
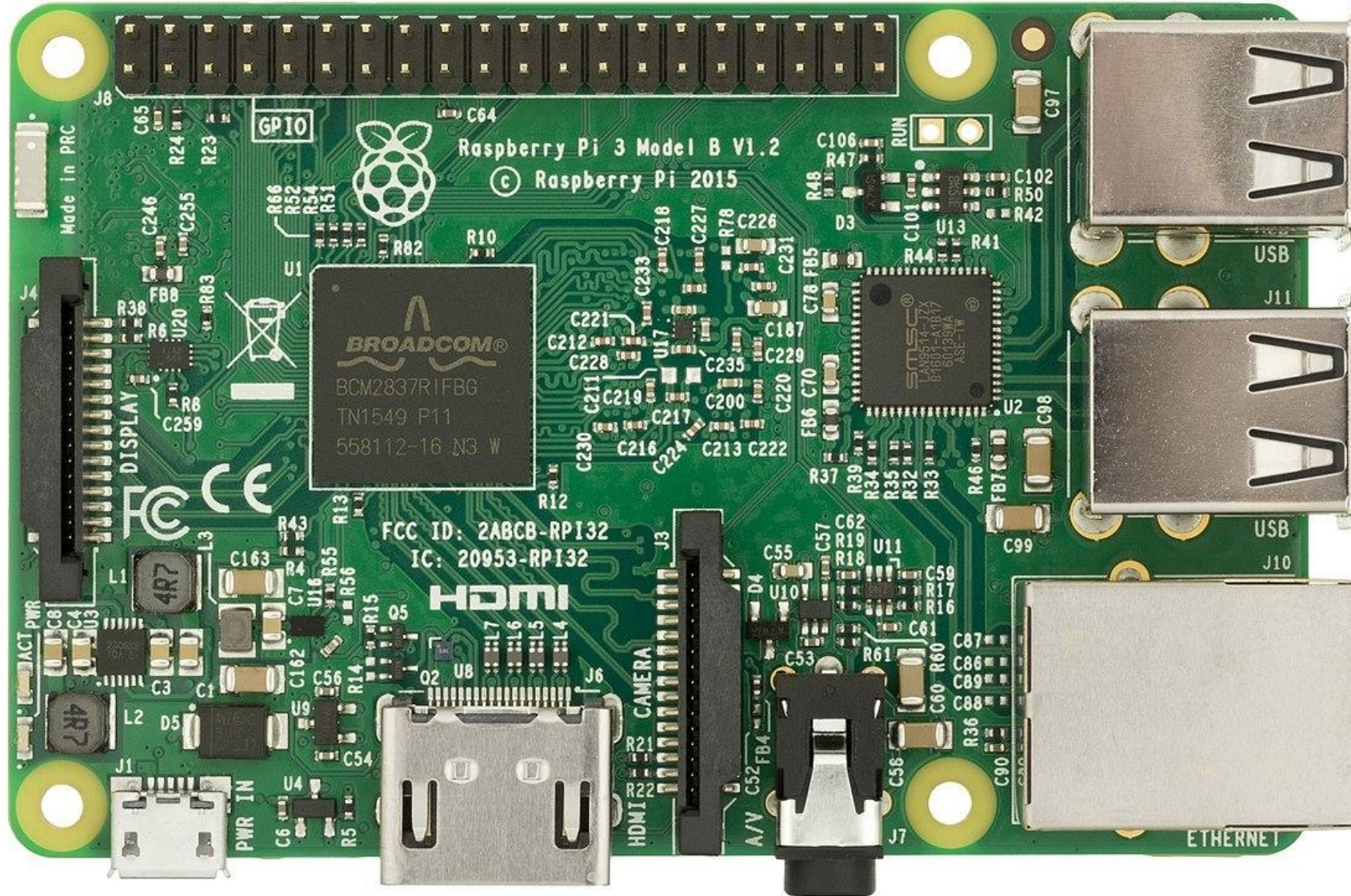


Fig : Raspberry pi

Raspberry Pi



Arduino vs. Raspberry Pi

- ❖ Raspberry Pi is a more complex system, much like a minicomputer. It has an advanced processor, uses Linux-based operating systems, and is able to perform multiple tasks with complex computation.
- ❖ Raspberry Pi gives the option for Wi-Fi connectivity and access point formation, something which would require external components to be connected on to a regular Arduino board.
- ❖ The choice of selecting Raspberry Pi or Arduino depends solely on one's application.
- ❖ If users want to perform rudimentary, basic level tasks then Arduino is the best option; it avoids any kind of complexities. If the intended application needs multiple tasks to be done with complex computations, Raspberry Pi is a much more suitable choice as it has pretty much everything integrated on-board.

Arduino vs. Raspberry Pi

- ❖ An Arduino board has a mounted micro controller chip to perform simple tasks. The board can execute single tasks at a time, in a repetitive manner. The main focus area of Arduino is integration of hardware to the board rather than complex computation or multitasking.
- ❖ For example, the Arduino board can sense the ambient temperature of an area and display it on an output console.
- ❖ The choice of selecting Raspberry Pi or Arduino depends solely on one's application.
- ❖ If users want to perform rudimentary, basic level tasks then Arduino is the best option; it avoids any kind of complexities. If the intended application needs multiple tasks to be done with complex computations, Raspberry Pi is a much more suitable choice as it has pretty much everything integrated on-board.

Latest Raspberry Models

Family ♦	Model ♦	SoC ♦	Memory ♦	Form factor ♦	Ethernet ♦	Wireless ♦	GPIO ♦	Released ♦	Discontinued
Raspberry Pi 4	400	BCM2711B0 / C0 ^[56]	4 GB	Keyboard	Yes ^[g]	Yes ^[e]	40-pin	2020	No
Raspberry Pi Pico	Pico	RP2040	264 KB	Pico ^[h]	No	No	40-pin	2021	No
Raspberry Pi Pico	W	RP2040	264 KB	Pico ^[h]	No	Yes ^[i]	40-pin	2022	No
Raspberry Pi Pico	2	RP2350A	520 KB	Pico ^[h]	No	No	40-pin	2024	No
Raspberry Pi Pico	2 W	RP2350A	520 KB	Pico ^[h]	No	Yes ^[i]	40-pin	2024	No
Raspberry Pi 5 ^[60]	B	BCM2712	2 GB	Standard ^[a]	Yes ^[g]	Yes ^[e]	40-pin	2024	No
Raspberry Pi 5 ^[60]	B	BCM2712	4 GB	Standard ^[a]	Yes ^[g]	Yes ^[e]	40-pin	2023	No
Raspberry Pi 5 ^[60]	B	BCM2712	8 GB	Standard ^[a]	Yes ^[g]	Yes ^[e]	40-pin	2023	No
Raspberry Pi 5 ^[60]	B	BCM2712	16 GB	Standard ^[a]	Yes ^[g]	Yes ^[e]	40-pin	2025 ^[61]	No
Raspberry Pi 5 ^[60]	500	BCM2712	8 GB	Keyboard	Yes ^[g]	Yes ^[e]	40-pin	2024 ^[62]	No

♦ The Broadcom BCM2712 is an entry-level ARM SoC with a quad-core ARM Cortex-A76 processor

Thank You